

TASK 3: STORYBOARD

Product: Samsung AI Family Hub Smart Refrigerator

Story Title:

A Smarter Way to Manage Groceries and Daily Kitchen Tasks

Scene 1: Realizing the Problem

Swati is preparing breakfast in the morning when she notices that the milk and eggs are finished. She feels frustrated because she forgot to check her grocery stock before.

Emotion: Frustrated

Scene 2: Checking Refrigerator Inventory

Instead of manually checking every shelf, Sneha opens the Samsung Family Hub application on her smartphone. The refrigerator instantly shows the available food items stored inside.

Emotion: Curious

Scene 3: Smart Food Management

The refrigerator automatically detects items that are running low and highlights products that are close to their expiry date. This helps Swati understand what needs to be used first.

Emotion: Relieved

Scene 4: Creating a Shopping List

The AI-powered system suggests adding milk, eggs, and vegetables to the shopping list. Swati does not have to remember everything herself.

Emotion: Comfortable

Scene 5: Personalized Recommendations

Based on the available ingredients, the refrigerator recommends recipes that can be prepared at home. Swati explores the suggestions and plans dinner accordingly.

Emotion: Excited

Scene 6: Improved User Experience

After shopping and preparing meals, Sneha realizes that the Samsung AI Family Hub Refrigerator has made grocery management easier, reduced food waste, and saved her valuable time.

Emotion: Satisfied

Storyboard Flow

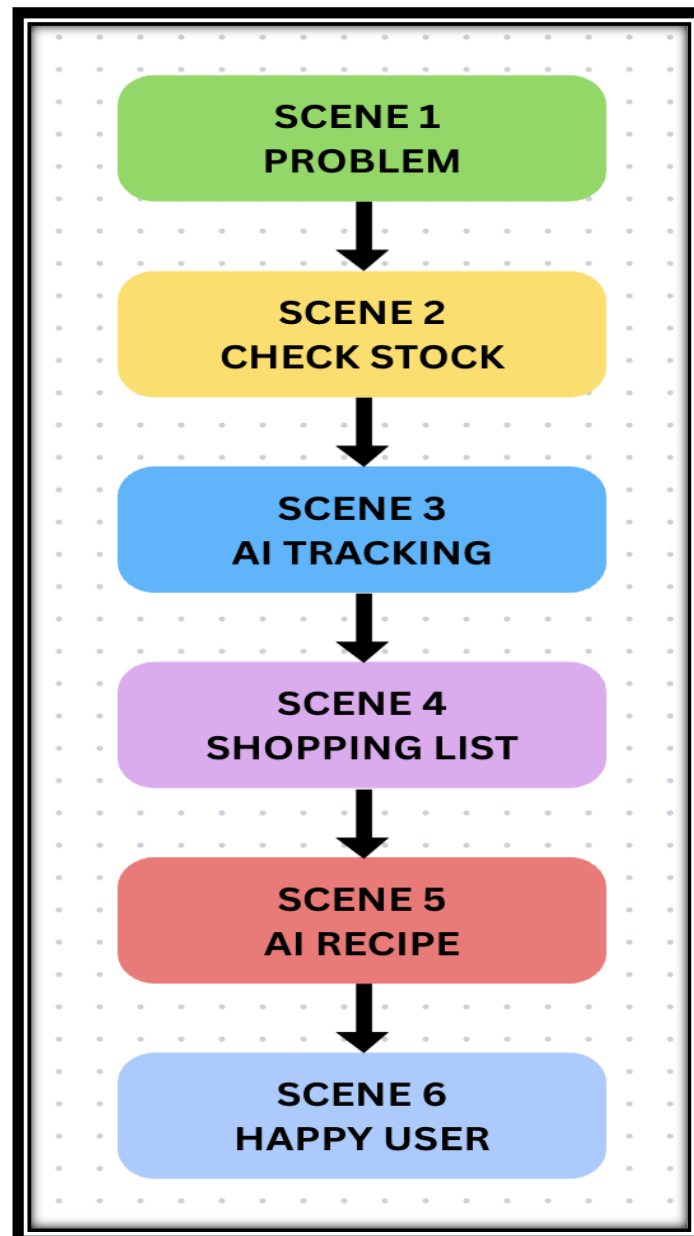


FIGURE 3.1

TASK 4: EMPATHY MAP & DESIGN THINKING APPROACH

Product: Smart Trash Bin

Problem Statement

The Smart Trash Bin is not being widely adopted because users face difficulties with unreliable sensors and confusing LED indicators. As a result, many users prefer traditional trash bins over the smart solution.

EMPATHY MAP

Says

- "The bin doesn't open every time I wave my hand."
- "I don't understand what these LED colors mean."
- "Using a normal dustbin seems easier."
- "This technology is confusing."

Thinks

- The product should be simpler to use.
- Maybe the sensor is not working properly.
- I expected a better experience for the price.
- Traditional bins require less effort.

Feels

- Frustrated when the sensor fails.
- Confused about LED indicators.
- Disappointed with the user experience.
- Annoyed when the bin does not respond quickly.

Does

- Waves hand multiple times near the sensor.
- Tries to open the bin manually.
- Avoids using smart features.
- Shares negative feedback with others.

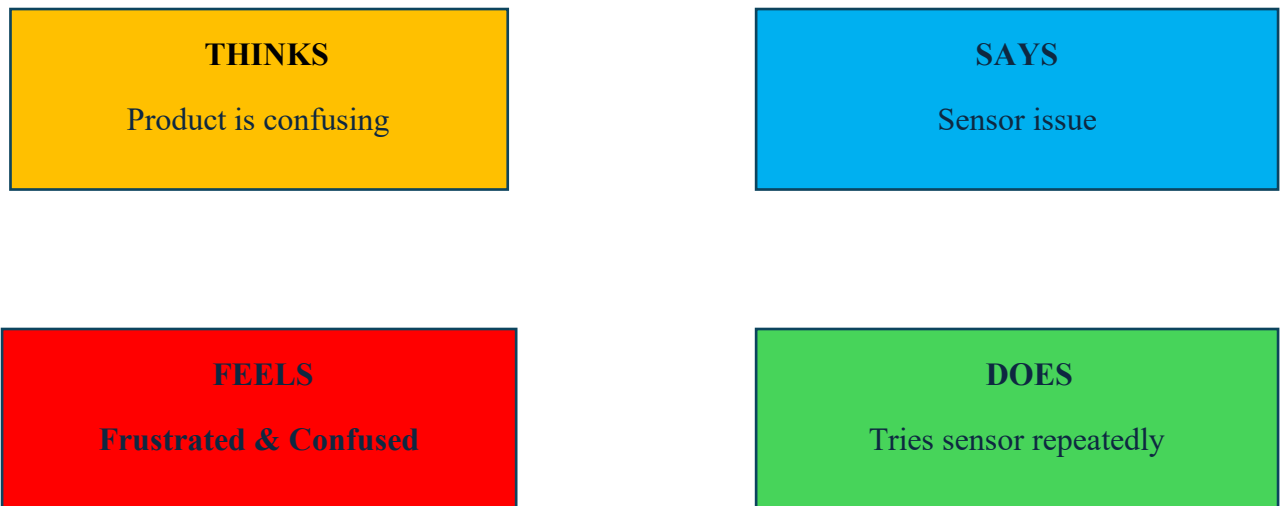


Figure 4.1

DESIGN THINKING APPROACH

1. Empathize

Observe users while using the Smart Trash Bin and understand their frustrations.

Findings:

- Sensor does not respond consistently.
- LED indicators are difficult to understand.
- Users need a simpler and more intuitive experience.

2. Define

Problem Statement

Users find the Smart Trash Bin difficult to use because the sensor is unreliable and the LED indicators do not clearly communicate the bin's status.

3. Ideate

Possible Solutions

- Improve sensor accuracy and response time.
- Use color-coded LED indicators:
 - ● Green = Working Normally
 - ● Yellow = Low Battery
 - ● Red = Error
- Add voice guidance for users.
- Provide a quick-start guide inside the package.
- Create simple onboarding instructions through a mobile app.

4. Prototype

Develop a new version of the Smart Trash Bin with:

- Advanced motion sensors
- Clear LED indicators
- Voice notification feature
- Easy-to-understand user interface

5. Test

Test the prototype with different users and collect feedback.

Evaluation Criteria

- Sensor responsiveness
- Ease of understanding LED indicators
- Overall user satisfaction

Based on user feedback, further improvements can be made before launching the final product.